Structure of the linearized model for the wind farm model

$$
\left(\begin{array}{cccc|cccc|cccc|cccc|ccc|ccc|ccc}
-a^{px}_{3,2} & a^{nx}_{3,2} & 0 & 0 & a^{ex}_{3,2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
a^{sx}_{3,3} & -a^{px}_{3,3} & a^{nx}_{3,3} & 0 & 0 & a^{ex}_{3,3} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & a^{sx}_{3,4} & -a^{px}_{3,4} & a^{nx}_{3,4} & 0 & 0 & a^{ex}_{3,4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & a^{sx}_{3,5} & -a^{px}_{3,5} & 0 & 0 & 0 & a^{ex}_{3,5} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\hline
a^{wx}_{4,2} & 0 & 0 & 0 & -a^{px}_{4,2} & a^{nx}_{4,2} & 0 & 0 & a^{ex}_{4,2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & a^{wx}_{4,3} & 0 & 0 & a^{sx}_{4,3} & -a^{px}_{4,3} & a^{nx}_{4,3} & 0 & 0 & a^{ex}_{4,3} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & a^{wx}_{4,4} & 0 & 0 & a^{sx}_{4,4} & -a^{px}_{4,4} & a^{nx}_{4,4} & 0 & 0 & a^{ex}_{4,4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & a^{wx}_{4,5} & 0 & 0 & a^{sx}_{4,5} & -a^{px}_{4,5} & 0 & 0 & 0 & a^{ex}_{4,5} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\hline
0 & 0 & 0 & 0 & a^{wx}_{5,2} & 0 & 0 & 0 & -a^{px}_{5,2} & a^{nx}_{5,2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & a^{wx}_{5,3} & 0 & 0 & a^{sx}_{5,3} & -a^{px}_{5,3} & a^{nx}_{5,3} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & a^{wx}_{5,4} & 0 & 0 & a^{sx}_{5,4} & -a^{px}_{5,4} & a^{nx}_{5,4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & a^{wx}_{5,5} & 0 & 0 & a^{sx}_{5,5} & -a^{px}_{5,5} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\hline
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -a^{py}_{2,3} & a^{ny}_{2,3} & 0 & a^{ey}_{2,3} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & a^{sy}_{2,4} & -a^{py}_{2,4} & a^{ny}_{2,4} & 0 & a^{ey}_{2,4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & a^{sy}_{2,5} & -a^{py}_{2,5} & 0 & 0 & a^{ey}_{2,5} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\hline
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & a^{wy}_{3,3} & 0 & 0 & -a^{py}_{3,3} & a^{ny}_{3,3} & 0 & a^{ey}_{3,3} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & a^{wy}_{3,4} & 0 & a^{sy}_{3,4} & -a^{py}_{3,4} & a^{ny}_{3,4} & 0 & a^{ey}_{3,4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & a^{wy}_{3,5} & 0 & a^{sy}_{3,5} & -a^{py}_{3,5} & 0 & 0 & a^{ey}_{3,5} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\hline
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & a^{wy}_{4,3} & 0 & 0 & -a^{py}_{4,3} & a^{ny}_{4,3} & 0 & a^{ey}_{4,3} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & a^{wy}_{4,4} & 0 & a^{sy}_{4,4} & -a^{py}_{4,4} & a^{ny}_{4,4} & 0 & a^{ey}_{4,4} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & a^{wy}_{4,5} & 0 & a^{sy}_{4,5} & -a^{Py}_{4,5} & 0 & 0 & a^{ey}_{4,5} & 0 & 0 & 0 & 0 \\
\hline
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & a^{wy}_{5,3} & 0 & 0 & -a^{py}_{5,3} & a^{ny}_{5,3} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & a^{wy}_{5,4} & 0 & a^{sy}_{5,4} & -a^{py}_{5,4} & a^{ny}_{5,4} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & a^{wy}_{5,5} & 0 & a^{sy}_{5,5} & -a^{py}_{5,5} & 0 & 0 & 0 & 0
\end{array}\right)
\begin{bmatrix}
u_{3,2}\\ u_{3,3}\\ u_{3,4}\\ u_{3,5}\\ \hline
u_{4,2}\\ u_{4,3}\\ u_{4,4}\\ u_{4,5}\\ \hline
u_{5,2}\\ u_{5,3}\\ u_{5,4}\\ u_{5,5}\\ \hline
v_{2,3}\\ v_{2,4}\\ v_{2,5}\\ \hline
v_{3,3}\\ v_{3,4}\\ v_{3,5}\\ \hline
v_{4,3}\\ v_{4,4}\\ v_{4,5}\\ \hline
v_{5,3}\\ v_{5,4}\\ v_{5,5}
\end{bmatrix}
+ 2\rho A
\begin{bmatrix}
0\\ u^2_{3,3}\beta_1\\ u^2_{3,4}\beta_1\\ 0\\ \hline
0\\ 0\\ 0\\ 0\\ \hline
0\\ u^2_{5,3}\beta_2\\ u^2_{5,4}\beta_2\\ 0\\ \hline
0\\ 0\\ 0\\ \hline
0\\ 0\\ 0\\ \hline
0\\ 0\\ 0\\ \hline
0\\ 0\\ 0
\end{bmatrix}
=
\begin{bmatrix}
-a^{sx}_{3,2}u_{3,1} - a^{wx}_{3,2}u_{2,2}\\
-a^{wx}_{3,3}u_{2,3}\\
-a^{wx}_{3,4}u_{2,4}\\
-a^{wx}_{3,5}u_{2,5} - a^{nx}_{3,5}u_{3,6}\\ \hline
-a^{sx}_{4,2}u_{4,1}\\
0\\
0\\
-a^{nx}_{4,5}u_{4,6}\\ \hline
-a^{sx}_{5,2}u_{5,1} - a^{ex}_{5,2}u_{6,2}\\
-a^{ex}_{5,3}u_{6,3}\\
-a^{ex}_{5,4}u_{6,4}\\
-a^{nx}_{5,5}u_{5,6} - a^{ex}_{5,5}u_{6,5}\\ \hline
-a^{sy}_{2,3}v_{2,2} - a^{wy}_{2,3}v_{1,3}\\
-a^{wy}_{2,4}v_{1,4}\\
-a^{wy}_{2,5}v_{1,5} - a^{ny}_{2,5}v_{2,6}\\ \hline
-a^{sy}_{3,3}v_{3,2}\\
0\\
-a^{ny}_{3,5}v_{3,6}\\ \hline
-a^{sy}_{4,3}v_{4,2}\\
0\\
-a^{ny}_{4,5}v_{4,6}\\ \hline
-a^{ey}_{5,3}v_{6,3} - a^{sy}_{5,3}v_{5,2}\\
-a^{ey}_{5,4}v_{6,4}\\
-a^{ey}_{5,5}v_{6,5} - a^{ny}_{5,5}v_{5,6}
\end{bmatrix}
\tag{1}
$$

$$
\left(\begin{array}{cccc|cccc|cccc|cccc|ccc|ccc|ccc}
-\delta a^{px}_{3,2} & \delta a^{nx}_{3,2} & 0 & 0 & \delta a^{ex}_{3,2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \delta a^{nwx}_{3,2} & 0 & 0 & \delta a^{nex}_{3,2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\delta a^{sx}_{3,3} & -\delta a^{px}_{3,3} & \delta a^{nx}_{3,3} & 0 & 0 & \delta a^{ex}_{3,3} & 0 & 0 & 0 & 0 & 0 & 0 & \delta a^{swx}_{3,3} & \delta a^{nwx}_{3,3} & 0 & \delta a^{sex}_{3,3} & \delta a^{nex}_{3,3} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \delta a^{sx}_{3,4} & -\delta a^{px}_{3,4} & \delta a^{nx}_{3,4} & 0 & 0 & \delta a^{ex}_{3,4} & 0 & 0 & 0 & 0 & 0 & 0 & \delta a^{swx}_{3,4} & \delta a^{nwx}_{3,4} & 0 & \delta a^{sex}_{3,4} & \delta a^{nex}_{3,4} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \delta a^{sx}_{3,5} & -\delta a^{px}_{3,5} & 0 & 0 & 0 & \delta a^{ex}_{3,5} & 0 & 0 & 0 & 0 & 0 & 0 & \delta a^{swx}_{3,5} & 0 & 0 & \delta a^{sex}_{3,5} & 0 & 0 & 0 & 0 & 0 & 0 \\
\hline
\delta a^{wx}_{4,2} & 0 & 0 & 0 & -\delta a^{px}_{4,2} & \delta a^{nx}_{4,2} & 0 & 0 & \delta a^{ex}_{4,2} & 0 & 0 & 0 & 0 & 0 & 0 & \delta a^{nwx}_{4,2} & 0 & 0 & \delta a^{nex}_{4,2} & 0 & 0 & 0 & 0 & 0 \\
0 & \delta a^{wx}_{4,3} & 0 & 0 & \delta a^{sx}_{4,3} & -\delta a^{px}_{4,3} & \delta a^{nx}_{4,3} & 0 & 0 & \delta a^{ex}_{4,3} & 0 & 0 & 0 & 0 & 0 & \delta a^{swx}_{4,3} & \delta a^{nwx}_{4,3} & 0 & \delta a^{sex}_{4,3} & \delta a^{nex}_{4,3} & 0 & 0 & 0 & 0 \\
0 & 0 & \delta a^{wx}_{4,4} & 0 & 0 & \delta a^{sx}_{4,4} & -\delta a^{px}_{4,4} & \delta a^{nx}_{4,4} & 0 & 0 & \delta a^{ex}_{4,4} & 0 & 0 & 0 & 0 & 0 & \delta a^{swx}_{4,4} & \delta a^{nwx}_{4,4} & 0 & \delta a^{sex}_{4,4} & \delta a^{nex}_{4,4} & 0 & 0 & 0 \\
0 & 0 & 0 & \delta a^{wx}_{4,5} & 0 & 0 & \delta a^{sx}_{4,5} & -\delta a^{px}_{4,5} & 0 & 0 & 0 & \delta a^{ex}_{4,5} & 0 & 0 & 0 & 0 & 0 & \delta a^{swx}_{4,5} & 0 & 0 & \delta a^{sex}_{4,5} & 0 & 0 & 0 \\
\hline
0 & 0 & 0 & 0 & \delta a^{wx}_{5,2} & 0 & 0 & 0 & -\delta a^{px}_{5,2} & \delta a^{nx}_{5,2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \delta a^{nwx}_{5,2} & 0 & 0 & \delta a^{nex}_{5,2} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \delta a^{wx}_{5,3} & 0 & 0 & \delta a^{sx}_{5,3} & -\delta a^{px}_{5,3} & \delta a^{nx}_{5,3} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \delta a^{swx}_{5,3} & \delta a^{nwx}_{5,3} & 0 & \delta a^{sex}_{5,3} & \delta a^{nex}_{5,3} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & \delta a^{wx}_{5,4} & 0 & 0 & \delta a^{sx}_{5,4} & -\delta a^{px}_{5,4} & \delta a^{nx}_{5,4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \delta a^{swx}_{5,4} & \delta a^{nwx}_{5,4} & 0 & \delta a^{sex}_{5,4} & \delta a^{nex}_{5,4} \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \delta a^{wx}_{5,5} & 0 & 0 & \delta a^{sx}_{5,5} & -\delta a^{px}_{5,5} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \delta a^{swx}_{5,5} & 0 & 0 & \delta a^{sex}_{5,5} \\
\hline
\delta a^{sey}_{2,3} & \delta a^{ney}_{2,3} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\delta a^{py}_{2,3} & \delta a^{ny}_{2,3} & 0 & \delta a^{ey}_{2,3} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \delta a^{sey}_{2,4} & \delta a^{ney}_{2,4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \delta a^{sy}_{2,4} & -\delta a^{py}_{2,4} & \delta a^{ny}_{2,4} & 0 & \delta a^{ey}_{2,4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \delta a^{sey}_{2,5} & \delta a^{ney}_{2,5} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \delta a^{sy}_{2,5} & -\delta a^{py}_{2,5} & 0 & 0 & \delta a^{ey}_{2,5} & 0 & 0 & 0 & 0 & 0 & 0 \\
\hline
\delta a^{swy}_{3,3} & \delta a^{nwy}_{3,3} & 0 & 0 & \delta a^{sey}_{3,3} & \delta a^{ney}_{3,3} & 0 & 0 & 0 & 0 & 0 & 0 & \delta a^{wy}_{3,3} & 0 & 0 & -\delta a^{py}_{3,3} & \delta a^{ny}_{3,3} & 0 & \delta a^{ey}_{3,3} & 0 & 0 & 0 & 0 & 0 \\
0 & \delta a^{swy}_{3,4} & \delta a^{nwy}_{3,4} & 0 & 0 & \delta a^{sey}_{3,4} & \delta a^{ney}_{3,4} & 0 & 0 & 0 & 0 & 0 & 0 & \delta a^{wy}_{3,4} & 0 & \delta a^{sy}_{3,4} & -\delta a^{py}_{3,4} & \delta a^{ny}_{3,4} & 0 & \delta a^{ey}_{3,4} & 0 & 0 & 0 & 0 \\
0 & 0 & \delta a^{swy}_{3,5} & \delta a^{nwy}_{3,5} & 0 & 0 & \delta a^{sey}_{3,5} & \delta a^{ney}_{3,5} & 0 & 0 & 0 & 0 & 0 & 0 & \delta a^{wy}_{3,5} & 0 & \delta a^{sy}_{3,5} & -\delta a^{py}_{3,5} & 0 & 0 & \delta a^{ey}_{3,5} & 0 & 0 & 0 \\
\hline
0 & 0 & 0 & 0 & \delta a^{swy}_{4,3} & \delta a^{nwy}_{4,3} & 0 & 0 & \delta a^{sey}_{4,3} & \delta a^{ney}_{4,3} & 0 & 0 & 0 & 0 & 0 & \delta a^{wy}_{4,3} & 0 & 0 & -\delta a^{py}_{4,3} & \delta a^{ny}_{4,3} & 0 & \delta a^{ey}_{4,3} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \delta a^{swy}_{4,4} & \delta a^{nwy}_{4,4} & 0 & 0 & \delta a^{sey}_{4,4} & \delta a^{ney}_{4,4} & 0 & 0 & 0 & 0 & 0 & \delta a^{wy}_{4,4} & 0 & \delta a^{sy}_{4,4} & -\delta a^{py}_{4,4} & \delta a^{ny}_{4,4} & 0 & \delta a^{ey}_{4,4} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & \delta a^{swy}_{4,5} & \delta a^{nwy}_{4,5} & 0 & 0 & \delta a^{sey}_{4,5} & \delta a^{ney}_{4,5} & 0 & 0 & 0 & 0 & 0 & \delta a^{wy}_{4,5} & 0 & \delta a^{sy}_{4,5} & -\delta a^{py}_{4,5} & 0 & 0 & \delta a^{ey}_{4,5} \\
\hline
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \delta a^{swy}_{5,3} & \delta a^{nwy}_{5,3} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \delta a^{wy}_{5,3} & 0 & 0 & -\delta a^{py}_{5,3} & \delta a^{ny}_{5,3} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \delta a^{swy}_{5,4} & \delta a^{nwy}_{5,4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \delta a^{wy}_{5,4} & 0 & \delta a^{sy}_{5,4} & -\delta a^{py}_{5,4} & \delta a^{ny}_{5,4} \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \delta a^{swy}_{5,5} & \delta a^{nwy}_{5,5} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \delta a^{wy}_{5,5} & 0 & \delta a^{sy}_{5,5} & -\delta a^{py}_{5,5}
\end{array}\right)
\begin{bmatrix}
\delta u_{3,2}\\ \delta u_{3,3}\\ \delta u_{3,4}\\ \delta u_{3,5}\\ \hline
\delta u_{4,2}\\ \delta u_{4,3}\\ \delta u_{4,4}\\ \delta u_{4,5}\\ \hline
\delta u_{5,2}\\ \delta u_{5,3}\\ \delta u_{5,4}\\ \delta u_{5,5}\\ \hline
\delta v_{2,3}\\ \delta v_{2,4}\\ \delta v_{2,5}\\ \hline
\delta v_{3,3}\\ \delta v_{3,4}\\ \delta v_{3,5}\\ \hline
\delta v_{4,3}\\ \delta v_{4,4}\\ \delta v_{4,5}\\ \hline
\delta v_{5,3}\\ \delta v_{5,4}\\ \delta v_{5,5}
\end{bmatrix}
\tag{2}
$$

Smx

$$
\left(\begin{array}{cccc|cccc|cccc||cccc|cccc|cccc|cccc}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 4\rho A u_{3,3}\beta_1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 4\rho A u_{3,4}\beta_1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\hline
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\hline
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 4\rho A u_{5,3}\beta_2 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 4\rho A u_{5,4}\beta_2 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\hline\hline
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\hline
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\hline
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\hline
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0
\end{array}\right)
\begin{bmatrix}
\delta u_{3,2} \\ \delta u_{3,3} \\ \delta u_{3,4} \\ \delta u_{3,5} \\ \hline
\delta u_{4,2} \\ \delta u_{4,3} \\ \delta u_{4,4} \\ \delta u_{4,5} \\ \hline
\delta u_{5,2} \\ \delta u_{5,3} \\ \delta u_{5,4} \\ \delta u_{5,5} \\ \hline\hline
\delta v_{2,3} \\ \delta v_{2,4} \\ \delta v_{2,5} \\ \hline
\delta v_{3,3} \\ \delta v_{3,4} \\ \delta v_{3,5} \\ \hline
\delta v_{4,3} \\ \delta v_{4,4} \\ \delta v_{4,5} \\ \hline
\delta v_{5,3} \\ \delta v_{5,4} \\ \delta v_{5,5}
\end{bmatrix} \tag{3}
$$

Boundary Conditions after update by previous solution

$$
\begin{bmatrix}
-a^{sx}_{3,2}(v_{2,3},v_{3,3})u_{3,2} - a^{wx}_{3,2}(u_{\infty},u_{3,2})u_{\infty} \\
-a^{wx}_{3,3}(u_{\infty},u_{3,3})u_{\infty} \\
-a^{wx}_{3,4}(u_{\infty},u_{3,4})u_{\infty} \\
-a^{wx}_{3,5}(u_{\infty},u_{3,5})u_{\infty} - a^{nx}_{3,5}(v_{2,5},v_{3,5})u_{3,5} \\ \hline
-a^{sx}_{4,2}(v_{3,3},v_{4,3})u_{4,2} \\
0 \\
0 \\
-a^{nx}_{4,5}(v_{3,5},v_{4,5})u_{4,5} \\ \hline
-a^{sx}_{5,2}(v_{4,3},v_{5,3})u_{5,2} - a^{ex}_{5,2}(2u_{5,2})u_{5,2} \\
-a^{ex}_{5,3}(2u_{5,3})u_{5,3} \\
-a^{ex}_{5,4}(2u_{5,4})u_{5,4} \\
-a^{nx}_{5,5}(v_{4,5},v_{5,5})u_{5,5} - a^{ex}_{5,5}(2u_{5,5})u_{5,5} \\ \hline\hline
-a^{sy}_{2,3}(2v_{2,3})v_{2,3} - a^{wy}_{2,3}(2u_{\infty})v_{\infty} \\
-a^{wy}_{2,4}(2u_{\infty})v_{\infty} \\
-a^{wy}_{2,5}(2u_{\infty})v_{\infty} - a^{ny}_{2,5}(2v_{2,5})v_{2,5} \\ \hline
-a^{sy}_{3,3}(2v_{3,3})v_{3,3} \\
0 \\
-a^{ny}_{3,5}(2v_{3,5})v_{3,5} \\ \hline
-a^{sy}_{4,3}(2v_{4,3})v_{4,3} \\
0 \\
-a^{ny}_{4,5}(2v_{4,5})v_{4,5} \\ \hline
-a^{ey}_{5,3}(u_{5,2},u_{5,3})v_{5,3} - a^{sy}_{5,3}(2v_{5,3})v_{5,3} \\
-a^{ey}_{5,4}(u_{5,3},u_{5,4})v_{5,4} \\
-a^{ey}_{5,5}(u_{5,4},u_{5,5})v_{5,5} - a^{ny}_{5,5}(2v_{5,5})v_{5,5}
\end{bmatrix} \tag{4}
$$

state variables affecting boundary conditions in the next iteration?????

$$
\begin{bmatrix}
\delta u_{3,2}\\ \delta u_{3,3}\\ \delta u_{3,4}\\ \delta u_{3,5}\\ \hline
\delta u_{4,2}\\ \vdots\\ \delta u_{4,5}\\ \hline
\delta u_{5,2}\\ \delta u_{5,3}\\ \delta u_{5,4}\\ \delta u_{5,5}\\ \hline\hline
\delta v_{2,3}\\ \vdots\\ \delta v_{2,5}\\ \hline
\delta v_{3,3}\\ \vdots\\ \delta v_{3,5}\\ \hline
\delta v_{4,3}\\ \vdots\\ \delta v_{4,5}\\ \hline
\delta v_{5,3}\\ \delta v_{5,4}\\ \delta v_{5,5}
\end{bmatrix}
\tag{5}
$$

$$
\left(\begin{array}{cccc|cccc|cccc||cccccccccccc}
-\delta a^{wx}_{3,2}u_\infty - a^{sx}_{3,2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\delta a^{sx}_{3,2}u_{3,2} & 0 & 0 & -\delta a^{sx}_{3,2}u_{3,2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
0 & -\delta a^{wx}_{3,3}u_\infty & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
0 & 0 & -\delta a^{wx}_{3,4}u_\infty & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
0 & 0 & 0 & -\delta a^{wx}_{3,5}u_\infty - a^{nx}_{3,5} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\delta a^{nx}_{3,5}u_{3,5} & 0 & 0 & -\delta a^{nx}_{3,5}u_{3,5} & 0 & 0 & 0 & 0 & 0 & 0\\
\hline
0 & 0 & 0 & 0 & -a^{sx}_{4,2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\delta a^{sx}_{4,2}u_{4,2} & 0 & 0 & -\delta a^{sx}_{4,2}u_{4,2} & 0 & 0 & 0 & 0 & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -a^{nx}_{4,5} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\delta a^{nx}_{4,5}u_{4,5} & 0 & 0 & -\delta a^{nx}_{4,5}u_{4,5} & 0 & 0 & 0\\
\hline
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -2\delta a^{ex}_{5,2}u_{5,2} - a^{ex}_{5,2} - a^{sx}_{5,2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\delta a^{sx}_{5,2}u_{5,2} & 0 & 0 & -\delta a^{sx}_{5,2}u_{5,2} & 0 & 0 & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -2\delta a^{ex}_{5,3}u_{5,3} - a^{ex}_{5,3} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -2\delta a^{ex}_{5,4}u_{5,4} - a^{ex}_{5,4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -2\delta a^{ex}_{5,5}u_{5,5} - a^{ex}_{5,5} - a^{nx}_{5,5} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\delta a^{nx}_{5,5}u_{5,5} & 0 & 0 & -\delta a^{nx}_{5,5}u_{5,5}\\
\hline\hline
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -2\delta a^{sy}_{2,3}v_{2,3} - a^{sy}_{2,3} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -2\delta a^{ny}_{2,5}v_{2,5} - a^{ny}_{2,5} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
\hline
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -2\delta a^{sy}_{3,3}v_{3,3} - a^{sy}_{3,3} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -2\delta a^{ny}_{3,5}v_{3,5} - a^{ny}_{3,5} & 0 & 0 & 0 & 0 & 0 & 0\\
\hline
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -2\delta a^{sy}_{4,3}v_{4,3} - a^{sy}_{4,3} & 0 & 0 & 0 & 0 & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -2\delta a^{ny}_{4,5}v_{4,5} - a^{ny}_{4,5} & 0 & 0 & 0 & 0\\
\hline
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\delta a^{ey}_{5,3}v_{5,3} & -\delta a^{ey}_{5,3}v_{5,3} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -2\delta a^{sy}_{5,3}v_{5,3} - a^{sy}_{5,3} - a^{ey}_{5,3} & 0 & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\delta a^{ey}_{5,4}v_{5,4} & -\delta a^{ey}_{5,4}v_{5,4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -a^{ey}_{5,4} & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\delta a^{ey}_{5,5}v_{5,5} & -\delta a^{ey}_{5,5}v_{5,5} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -2\delta a^{ny}_{5,5}v_{5,5} - a^{ny}_{5,5} - a^{ey}_{5,5}
\end{array}\right)
\begin{bmatrix}
\delta u_{3,2}\\ \delta u_{3,3}\\ \delta u_{3,4}\\ \delta u_{3,5}\\ \hline
\delta u_{4,2}\\ \delta u_{4,3}\\ \delta u_{4,4}\\ \delta u_{4,5}\\ \hline
\delta u_{5,2}\\ \delta u_{5,3}\\ \delta u_{5,4}\\ \delta u_{5,5}\\ \hline\hline
\delta v_{2,3}\\ \delta v_{2,4}\\ \delta v_{2,5}\\ \hline
\delta v_{3,3}\\ \delta v_{3,4}\\ \delta v_{3,5}\\ \hline
\delta v_{4,3}\\ \delta v_{4,4}\\ \delta v_{4,5}\\ \hline
\delta v_{5,3}\\ \delta v_{5,4}\\ \delta v_{5,5}
\end{bmatrix}
\tag{6}
$$